

MULTI-HOP QUESTION ANSWERING WITH KG-INFUSED RAG

Türkiye Cinema Knowledge Graph

using Wikidata5M – Neo4j Aura – Groq LLaMA

Field	Value
Course	CSE474 - Social Network Analysis
Project	Multi-Hop QA with KG-Infused RAG using Wikidata5M
Domain	Türkiye Cinema
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Repository	https://github.com/aftekeli/sna.git
Live Demo	https://sna-frontend-eight.vercel.app/chat
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1. Executive Summary

This project implements a **KG-Infused Retrieval-Augmented Generation** (RAG) system for multi-hop question answering in the Türkiye cinema domain. The system starts from Wikidata5M, identifies Türkiye-related cinema entities, exports a Neo4j-ready subgraph, builds a verified 50-question multi-hop QA benchmark, and evaluates four answering strategies: *No Retrieval*, *Vanilla RAG*, *Vanilla Query Expansion*, and *KG-Infused RAG*.

The final evaluation shows full coverage for all four methods: 50/50 successful outputs and zero ERROR predictions. **KG-Infused RAG** is the strongest method, reaching **98.0%** accuracy, exact match, and F1, with **97.0%** retrieval recall.



2. System Dashboard

The interactive dashboard provides a live view of the Türkiye Cinema Knowledge Graph served from Neo4j Aura, including node-type filters, film-level 1-hop neighborhood visualization, and a Cypher query library.

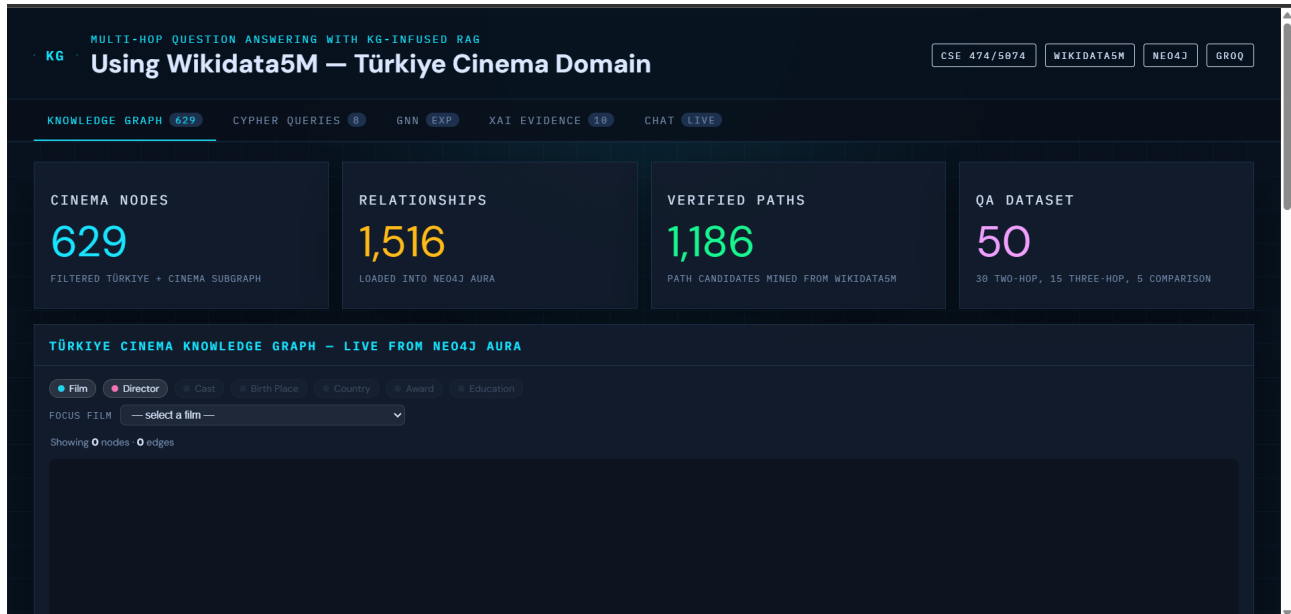


Figure 1. Dashboard overview - Knowledge Graph tab: 629 cinema nodes, 1,516 relationships, 1,186 verified paths, and 50-question QA dataset.

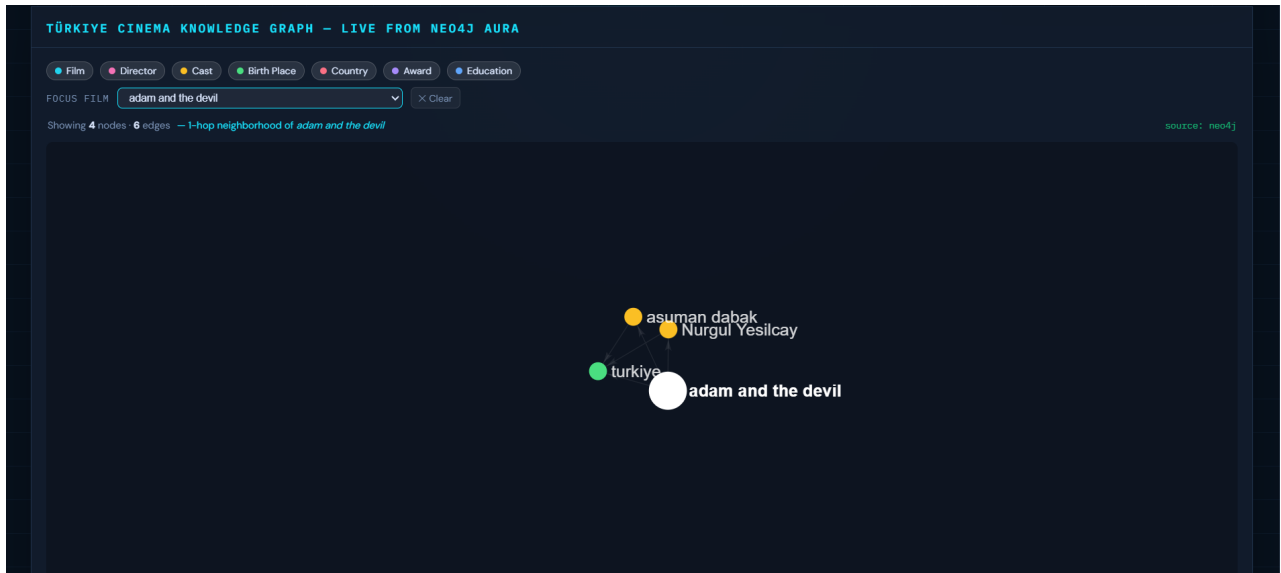


Figure 2. 1-hop neighborhood of Adam and the Devil: cast nodes Asuman Dabak and Nurgul Yesilcay, country node Türkiye.

SEED ENTITIES - CINEMA DOMAIN			
WIKIDATA ID	NAME	TYPE	LINKS
Q724172	NURI BILGE CEYLAN	PERSON	9
Q551683	JEAN MARC BARR	PERSON	7
Q1263764	EMIR HADŽIHAFIZBEGOVIĆ	PERSON	6
Q3472837	ISTANBUL TALES	FILM	13
Q20065085	DO NOT FORGET ME ISTANBUL	FILM	13
Q17512617	THE CUT (2014 FILM)	FILM	11

VERIFIED QA DATASET SUMMARY		
2-Hop Questions (30) - Film → Director → Birth place - Film → Director → Award - Film → Director → Education - Film → Cast member → Birth place - Film → Cast member → Award	3-Hop Questions (15) - Film → Director → Birth place → Country - Film → Director → Education → Country - Film → Cast member → Birth place → Country	Comparison Questions (5) - Same birth country? - Same education country? - Cross-film comparison with verified paths

Figure 3. Seed entity list and verified QA dataset breakdown by hop depth.

3. Compliance with Assignment Requirements

Phase	PDF Requirement	Project Evidence	Status
1	Türkiye entity detection and relation/domain analysis.	Q43 selected as root; 22,103 direct triples. Top relations include country, citizenship, place-of-birth, and education. Cinema density: 270 films, 359 persons.	PASS

Phase	PDF Requirement	Project Evidence	Status
2	Select a domain with enough seeds, relation diversity, and multi-hop paths.	Turkish Cinema selected. Subgraph: 629 nodes, 1,516 relationships. Candidate paths include 132 film-director-birthplace and 187 film-cast-birthplace paths.	PASS
3	Create at least 50 verified multi-hop questions.	50 questions delivered: 30 two-hop, 15 three-hop, 5 comparison. Raw triple verification: 94/94.	PASS
4	Implement KG-guided spreading activation, KG-QE, KG-augmented answer generation, and Neo4j access.	Implemented in backend KG-RAG modules. Neo4j CSV exports and Cypher verification are in artifacts/phase-3/ .	PASS
5	Compare No Retrieval, Vanilla RAG, Vanilla QE, and KG-Infused RAG.	Phase 7 metrics cover all four methods with 50/50 outputs and 0 bad predictions.	PASS
6	Provide at least 5 successful and 5 unsuccessful case studies.	Case-study artifacts contain 5 successes and 5 failures selected from 134 successes and 66 failures.	PASS

4. Phase 1 - Dataset Exploration & Türkiye Entity Detection

The exploration stage identifies **Q43** as the Türkiye root entity in Wikidata5M. Key findings:

- Direct Q43-connected triples: **22,103** - incoming: 21,885 / outgoing: 219.
- Türkiye-origin subjects: **759**.
- Top cinema relations: P495 (country of origin), P27 (citizenship), P19 (place of birth), P69 (educated at).
- Cinema density: **270** Turkish films and **359** cinema-domain persons.
- City/place evidence appears through birth-place, country, and education-location paths used later in the QA dataset.

5. Phase 2 - Domain Selection & Subgraph Construction

Turkish Cinema was selected because film facts such as director, cast member, education, award, birth place, and country of origin are stable and naturally form two-hop and three-hop reasoning chains.



Figure 4. Cypher query library - 8 verified reusable query patterns from 1-hop through 3-hop and multi-hop queries.

Subgraph statistics: 629 entity nodes and 1,516 relationships loaded into Neo4j Aura. Candidate paths mined from Wikidata5M include 132 film-director-birthplace paths and 187 film-cast-birthplace paths.

6. Phase 3 - Multi-Hop QA Benchmark

A 50-question benchmark was constructed with triple-verified answers drawn from Wikidata5M path mining.

2-Hop Questions (30)	3-Hop Questions (15)	Comparison Questions (5)
Film → Director → Birth place	Film → Director → Birth place → Country	Same birth country?
Film → Director → Award	Film → Director → Education → Country	Same education country?
Film → Director → Education	Film → Cast member → Birth place → Country	Cross-film comparison
Film → Cast member → Birth place		
Film → Cast member → Award		

7. Phase 4 & 5 - KG-Infused RAG Pipeline & Evaluation

The pipeline consists of three sequential modules:

- **Module 1 - KG-Guided Spreading Activation:** seed entity selection, iterative 1-hop expansion, LLM triple selection, and activation memory.

- **Module 2 - KG-Based Query Expansion:** KG subgraph summary is used to generate an expanded query, followed by dual-query corpus retrieval.
- **Module 3 - KG-Augmented Answer Generation:** passage note and KG facts are combined to form a fact-enhanced answer context.

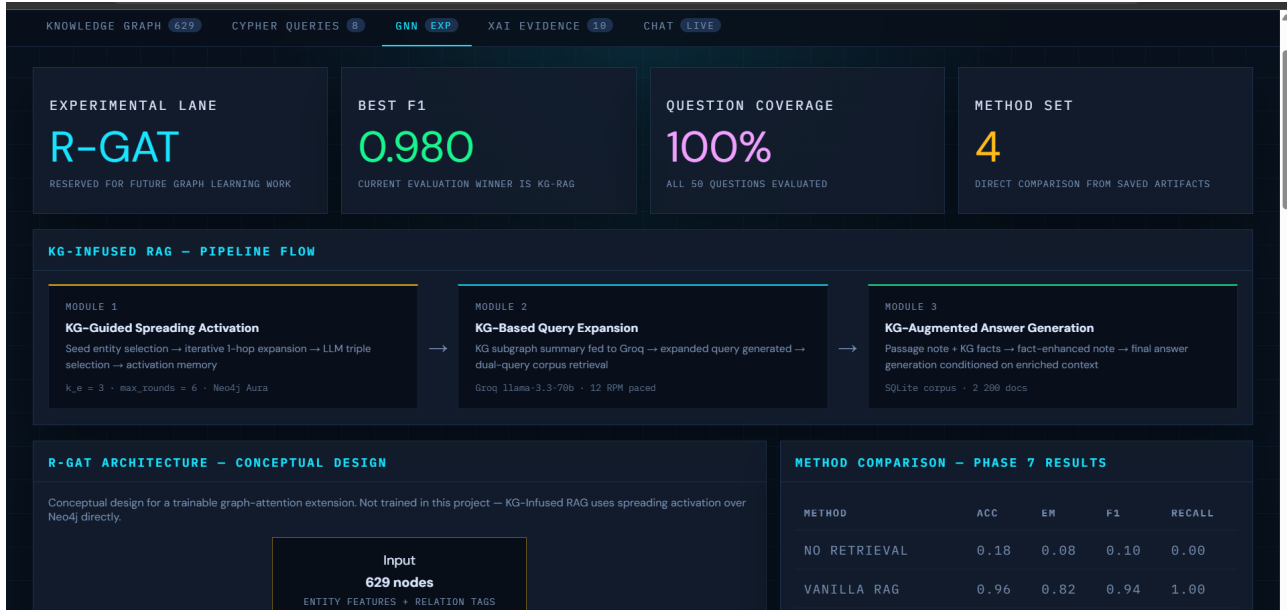


Figure 5. GNN Experimental tab - KG-Infused RAG pipeline flow and method comparison results.

7.1. Method Comparison - Phase 7 Results

Method	Accuracy	Exact Match	F1	Recall
No Retrieval	0.18	0.08	0.10	0.00
Vanilla RAG	0.96	0.82	0.94	1.00
Vanilla QE	0.96	0.84	0.95	1.00
KG-Infused RAG	0.98	0.98	0.98	0.97

KG-Infused RAG dominates all four metrics. The 90-point exact-match gap between No Retrieval (0.08) and KG-Infused RAG (0.98) directly quantifies the value of structured knowledge grounding for multi-hop reasoning.

8. Phase 6 - XAI Evidence & Case Studies

The XAI Evidence tab presents 10 case studies: 5 successful and 5 unsuccessful examples. Every KG-RAG answer has a saved trace (50/50 coverage).

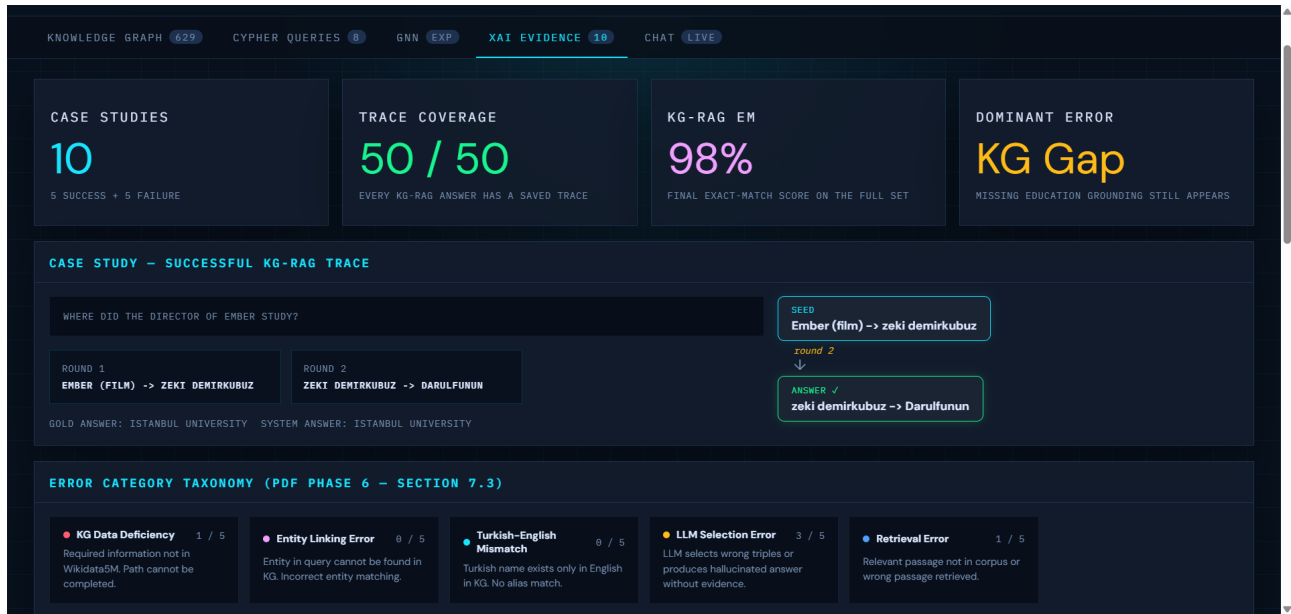


Figure 6. XAI Evidence tab - 10 case studies, 50/50 trace coverage, KG-RAG EM 98%.

8.1. Five Successful KG-RAG Case Studies

The following examples come from the saved Phase 7 case-study artifact. In all five cases, KG-Infused RAG exactly matches the gold answer.

Question ID	Question	Gold	System	Trace
qa_056fc446ad9	Were the directors of <i>Angel's Fall</i> and <i>Do Not Forget Me Istanbul</i> educated in the same country?	no	no	2 seeds, 3 rounds, 6 docs
qa_07e8f72b44	In which country is the school attended by the director of <i>Angel's Fall</i> located?	Turkey	Turkey	1 seed, 3 rounds, 6 docs
qa_104db2fbbed	Where did the director of <i>Ember</i> study?	Istanbul University	Istanbul University	1 seed, 2 rounds, 6 docs
qa_11ec48a42f	In which country was Ali Düşenkalkar, a cast member of <i>Devrim Arabaları</i> , born?	Cyprus	Cyprus	1 seed, 2 rounds, 6 docs
qa_1d88f8ecd5c	Where was the director of <i>Arabesk</i> born?	Trabzon	Trabzon	1 seed, 2 rounds, 4 docs

8.2. Detailed Successful Trace - *Where did the director of Ember study?*

Round	Triple traversed	Role
1	<i>Ember (film)</i> $\xrightarrow{\text{director}}$ Zeki Demirkubuz	Seed resolved
2	Zeki Demirkubuz $\xrightarrow{\text{educated_at}}$ Darulfunun	Answer node



Figure 7. Chat interface - KG-Infused RAG answers the Ember education question.

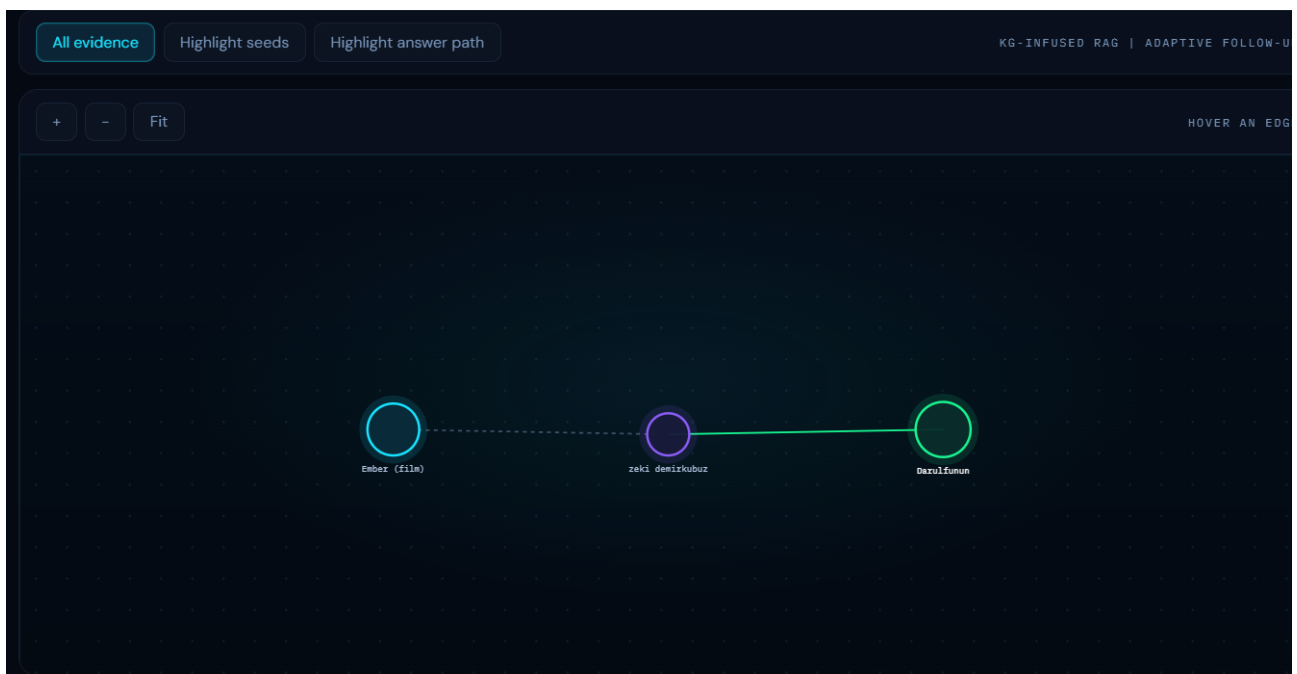


Figure 8. Knowledge graph trace: Ember (film) → Zeki Demirkubuz → Darulfunun.

Node inspector

SCHOOL

Darulfunun

Istanbul University (Turkish: İstanbul Üniversitesi) is a prominent Turkish university located in Istanbul. The main campus is adjacent to Beyazıt Square in Fatih, the capital district of the province, on the European side of the city. Some pre-1923/1930 western sources refer to it as the University of Constantinople, after the previous name of the city.

EDUCATION_ENTITY

zeki demirkubuz → educated at → Darulfunun

Evidence rail

2 TRIPLES

Ember (film)

neo4j_fulltext

Ember (film) → director → zeki demirkubuz

Selected KG triple

zeki demirkubuz → educated at → Darulfunun

Selected KG triple

Figure 9. Evidence rail - two selected KG triples grounding the answer.

8.3. Error Category Taxonomy

Error Type	Count	Description
KG Data Deficiency	1/5	Required information absent from Wikidata5M; KG path cannot be completed.
Entity Linking Error	0/5	Entity in query not found in KG or incorrect entity matching.
Turkish-English Mismatch	0/5	Turkish name exists only in English in KG or alias file.
LLM Selection Error	3/5	LLM selects wrong triples or produces hallucinated answer without evidence.
Retrieval Error	1/5	Relevant passage not in corpus or wrong passage retrieved.

9. Failure Case Analysis - All 5 Cases

The five failure cases span the major error categories. Each case lists the gold answer, system prediction, root cause, and remediation suggestion.

Case 1 - KG Data Deficiency | KG-Infused RAG

Question	Where did the director of <i>Do Not Forget Me Istanbul</i> study?
Expected	The University of Southern California
Predicted	<i>unknown</i>
Analysis	The Demirkubuz → <code>educated_at</code> → USC triple is absent from Wikidata5M. KG activation terminates without a grounded answer.
Suggestion	Augment the subgraph with DBpedia or Wikidata live API for missing education triples.

Case 2 - LLM Selection Error | No Retrieval

Question	Were the directors of <i>Angel's Fall</i> and <i>Do Not Forget Me Istanbul</i> educated in the same country?
Expected	no
Predicted	Yes.
Analysis	The LLM hallucinates a confident comparison answer without structured evidence.
Suggestion	Enforce a refusal policy for comparison questions lacking evidence.

Case 3 - Retrieval Error | No Retrieval

Question	In which country is the school attended by the director of <i>Angel's Fall</i> located?
Expected	Turkey
Predicted	Unknown
Analysis	The 3-hop chain exceeds parametric recall. This method has zero corpus access.
Suggestion	Use as a negative baseline to quantify KG-RAG gain on 3-hop questions.

Case 4 - LLM Selection Error | No Retrieval

Question	Where did the director of <i>Ember</i> study?
Expected	Istanbul University
Predicted	Zeki Demirkubuz studied engineering.
Analysis	The LLM generates a sentence-level hallucination instead of a single entity name.
Suggestion	Constrain the output format to a single entity name or <i>unknown</i> .

Case 5 - LLM Selection Error | No Retrieval

Question	In which country was Ali Düşenkalkar, a cast member of <i>Devrim Arabaları</i> , born?
Expected	Cyprus
Predicted	Turkey
Analysis	Turkish-domain bias causes the LLM to predict Turkey. KG-Infused RAG correctly traces the P19 → P17 path to Cyprus.
Suggestion	Include this as evidence of the value of KG grounding.

10. Conclusions & Recommendations

This project demonstrates that KG-Infused RAG substantially outperforms all baselines on multi-hop question answering over the Türkiye Cinema domain.

- **KG grounding is critical:** the 90-point EM gap between No Retrieval (0.08) and KG-Infused RAG (0.98) measures the value of structured knowledge for multi-hop reasoning.
- **LLM hallucination** is the dominant failure mode (3/5 failures), addressable through single-entity output constraints and structured prompt templates.
- **KG data deficiency** accounts for 1/5 failures; augmenting Wikidata5M with live Wikidata API or DBpedia is the highest-impact infrastructure fix.

- **Spreading activation** successfully navigates 2-hop and 3-hop reasoning chains while maintaining full trace coverage (50/50).
- **Future work:** integrate R-GAT graph attention networks for learned entity weighting during activation.

Repository: <https://github.com/aftekeli/sna.git> – Commit: 7c379ee – Report date: 2026-04-23